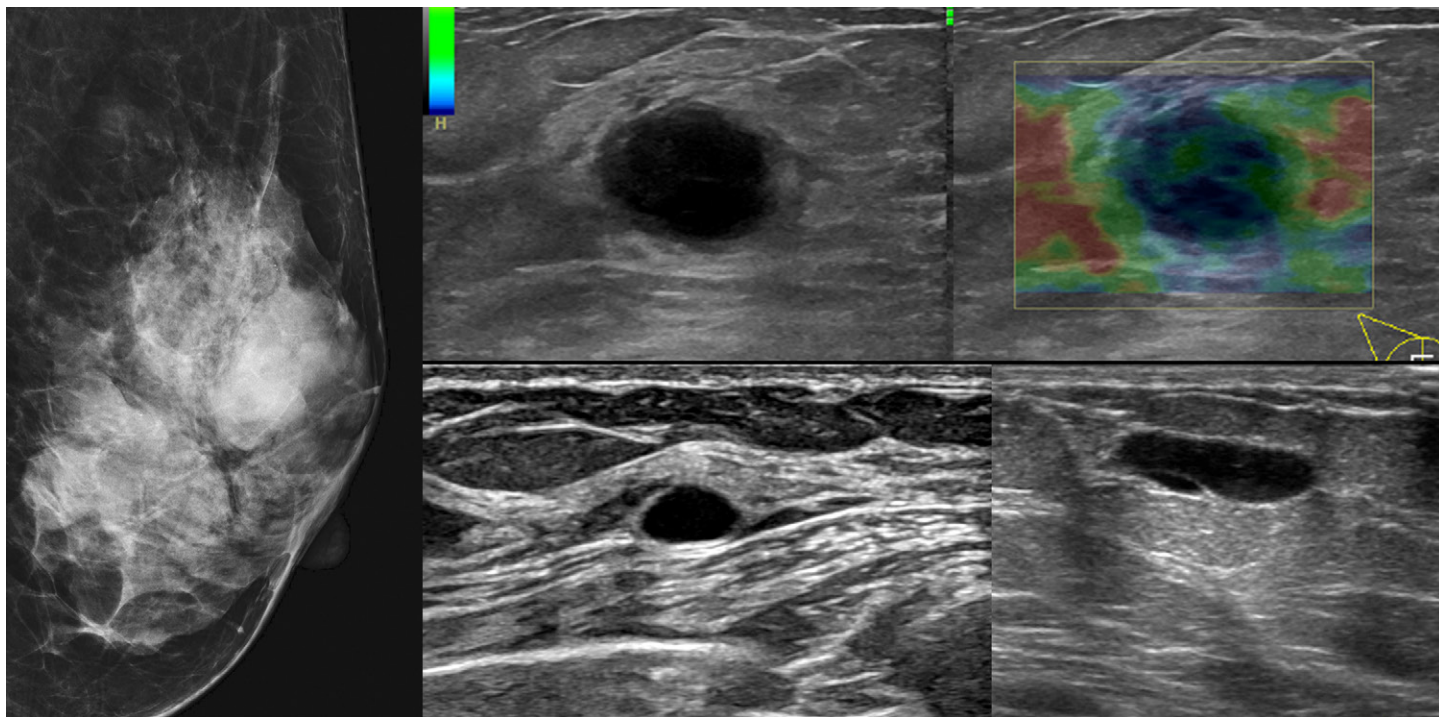


Cystic Breast Lesions: Diagnostic Approach and US Assessment

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Various cystic breast lesions are encountered during screening and diagnostic breast imaging. According to the Breast Imaging Reporting and Data System (BI-RADS) from the American College of Radiology, cystic breast lesions can be classified into the following categories based on sonographic findings: simple cysts, complicated cysts, clustered microcysts, and complex cystic and solid masses. With appropriate technique, simple cysts can be diagnosed easily by satisfying the diagnostic criteria, which include anechoic round or oval lesions with circumscribed margins and posterior enhancement on US images. Simple cysts are categorized as BI-RADS category 2, benign. Complicated cysts contain debris and satisfy all other sonographic criteria for simple cysts, except they are not anechoic. Clustered microcysts are defined as lesions comprising a cluster of small anechoic masses without a solid component. Based on recent investigations, complicated cysts are categorized as BI-RADS category 3, probably benign, whereas clustered microcysts are categorized as BI-RADS category 2. Complex cystic and solid masses contain fluid and solid components and include those with a thick wall, thick septations, an intracystic or mural mass, and both cystic and solid components. They usually are considered BI-RADS category 4, suspicious, and are accompanied by a biopsy recommendation. Radiologists must evaluate cystic lesions carefully, with meticulous technique, and provide appropriate assessment and management recommendations, thereby reducing unnecessary follow-up and biopsies while preventing cancers from being missed or dismissed.

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Introduction

Cystic breast lesions, especially simple cysts, are often observed at US performed for screening or diagnostic purposes and may be detected incidentally at breast MRI. These lesions include a wide spectrum of breast diseases, from common benign simple cysts and papillary lesions to breast carcino-

ma (1–4). Appropriate categorization by evaluating imaging features is important because clinical management of each lesion category is different. Consequently, there are studies that have attempted to classify and distinguish between benign and malignant cystic breast lesions by evaluating their sonographic features (1,2,5).

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Abbreviations: ACRIN = American College of Radiology Imaging Network, BI-RADS = Breast Imaging Reporting and Data System, H-E = hematoxylin-eosin

TEACHING POINTS

- Complicated cysts are masses that otherwise meet the sonographic criteria for simple cysts, except they are not anechoic on US images. A lesion easily can be diagnosed as a complicated cyst with debris and categorized as BI-RADS category 2 in cases of mobile internal echoes, a fluid-debris level, or echogenic reflector scintillating during real-time US image evaluation.
- BI-RADS category 2 is most appropriate for clustered microcysts that have a typical sonographic appearance. The presence of indistinct or angular margins, discrete solid components, associated suspicious calcifications at mammography, or a suspicious change at subsequent imaging should prompt a BI-RADS category 4 assessment and a recommendation for biopsy.
- Although echogenicity alone has low specificity in evaluation of a breast mass, complex cystic and solid masses usually are evaluated as BI-RADS category 4 and accompanied by a recommendation for biopsy.
- Simple cysts can be dismissed as benign (BI-RADS category 2) with a recommendation for routine screening. If the patient is symptomatic with pain or a very large cyst, therapeutic cyst aspiration can be performed, a seemingly discordant clinical scenario.
- Core-needle biopsy under US guidance is an effective method for pathologic diagnosis. It is important to obtain a sample of the solid components during biopsy because the cystic component could actually be necrosis.

The established terminology for cystic breast lesions, which was provided in the Breast Imaging Reporting and Data System (BI-RADS) from the American College of Radiology in 2013, classifies these lesions into the following four categories: (a) simple cysts, (b) complicated cysts, (c) clustered microcysts, and (d) complex cystic and solid masses (6). Simple cysts are the most common type. However, with the use of high-resolution US equipment, cysts with septations or internal echoes are also commonly encountered. Assessment and management recommendations for cystic breast lesions, especially for complicated cysts with debris and clustered microcysts, may be confusing, and verification through more studies has been needed. Several studies have recently reported the outcomes of nonsimple cysts seen at US, especially complicated cysts and clustered microcysts (7–10).

This article defines the four categories of cystic breast lesions; demonstrates imaging findings on mammographic, sonographic, and MR images; and discusses assessment and management recommendations from published literature.

Cystic Breast Lesions

Breast cysts are the most common nonproliferative disorder of the breast, with a peak incidence in patients between 35 and 50 years of age (4). Simple cysts are hormone sensitive and therefore have a peak incidence in premenopausal women.

In postmenopausal women, cysts are more common in those receiving hormone replacement therapy (4,11).

Simple cysts are epithelium-lined fluid-filled round or oval well-circumscribed masses. Several hypothetical mechanisms of cyst formation have been suggested. Cystic changes may be caused by obstruction of the terminal duct lobular unit, continuous secretion with impaired drainage, or alteration of lobules, starting with dilation and then coalescence of acini, with unfolding or involution of the lobular units. The epithelium lining the cysts can be flattened or apocrine metaplastic (12). Biologically active compounds secreted by the apocrine metaplastic epithelium increase the intraluminal pressure and cause adjacent acini to unfold and fuse, forming apocrine-lined microcysts. Smaller cysts gradually enlarge and fuse, forming larger cystic spaces (Fig 1) (12,13). Cysts with the apocrine cell lining have a higher risk for recurrence than those with flat epithelium (14).

It has been suggested that the natural history of cystic disease is closely related to the cyst type. Type I, a secretory cyst with a metabolically active apocrine cell lining, may have increased cell proliferation, with apocrine cells prone to atypia, hyperplasia, or preneoplastic alterations. Type II, a transudative cyst lined by flattened epithelium without hypertrophic metabolism, represents a mere retention cyst that is resolved after aspiration (12).

Evaluating Cystic Breast Lesions with US

US should be performed using a high-frequency transducer (ie, a broad-bandwidth linear-array transducer, with a center frequency of at least 10 MHz and a peak frequency of at least 12 MHz), with the appropriate gain and gray scale and focal zone at the level of the mass adjusted for each patient (4). Gain settings should be adjusted so that fat at all levels is displayed as a midlevel gray (15). One criterion for diagnosing a simple cyst is whether it is anechoic (Fig 2). Accurate gain adjustment, with optimal visualization of echogenicity, is important. Setting the gain too high can result in excessive echogenicity, which may obscure some lesions and make some cysts appear solid. Conversely, setting the gain too low may obscure details in the image, with hypoechoic solid lesions appearing anechoic.

Interpretation of B-mode images may be challenging because of artifactual echoes and difficulty in the evaluation of the lateral borders of breast lesions. Spatial compounding and harmonic imaging may facilitate evaluation of cystic breast lesions. Spatial compounding imaging is a technical approach in which the sonographic image is derived from the input of multiple off-angle beams (Fig S1). It results in decreased artifactual echoes and enhanced true echoes. Spatial compounding imaging also can aid in the evaluation of the lateral margins of the lesion. The improved signal-to-noise ratio assists in the recognition of small cysts. However, posterior enhancement is less apparent with spatial compounding, and the benefits decrease with increasing lesion depth (1,16).

Harmonic imaging, which uses higher harmonic frequencies that originate within tissues during beam propagation to produce gray-scale images, can be applied to better characterize cystic breast lesions (Fig S2). Lower-frequency superficial

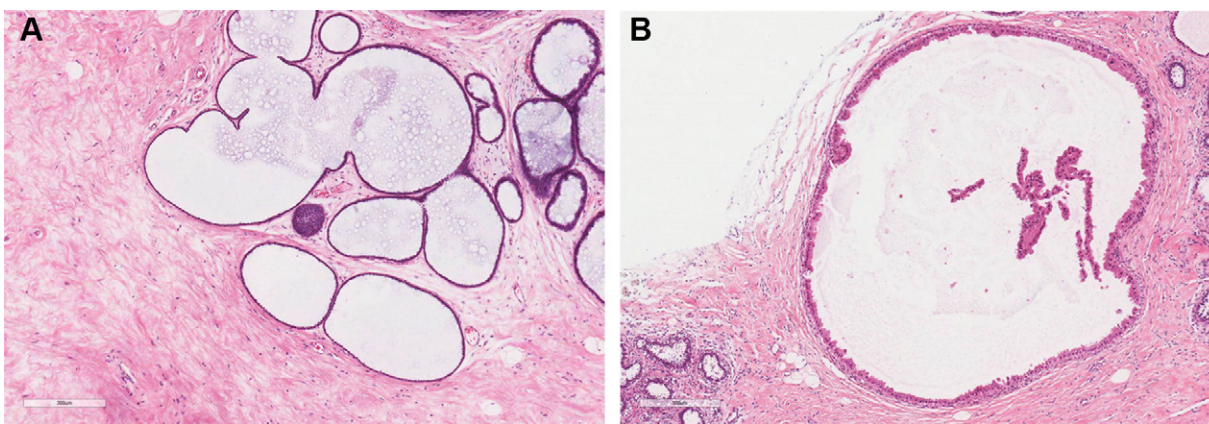


Figure 1. Cystic breast lesions. **(A)** Photomicrograph shows cysts lined by flattened epithelium. Adjacent small cysts have fused to form a larger cystic space. **(B)** Photomicrograph shows a cyst lined by apocrine metaplastic epithelium. (Hematoxylin-eosin [H-E] stain; original magnification, $\times 68$.)

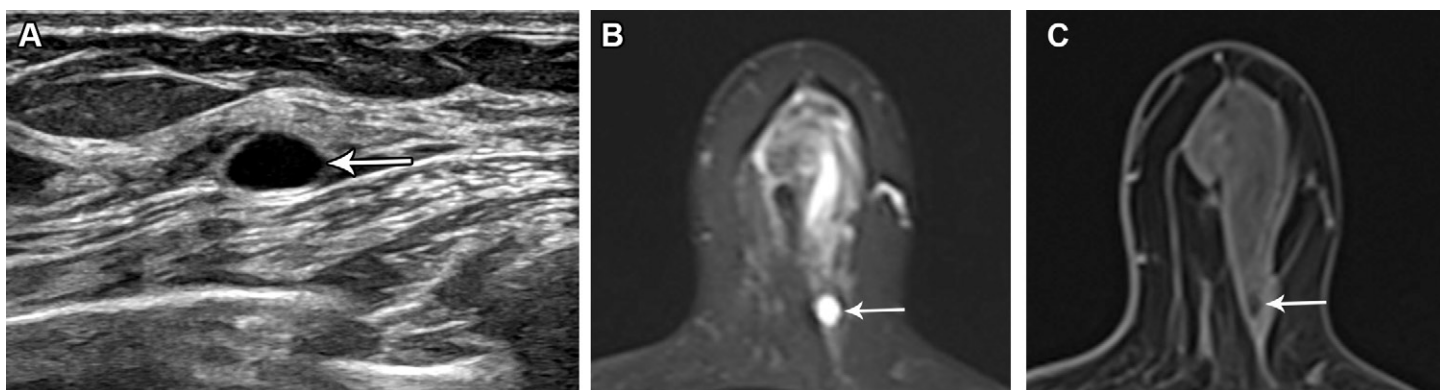


Figure 2. Simple cyst in a 62-year-old woman. **(A)** US image shows an oval circumscribed anechoic lesion (arrow) with posterior enhancement (ie, a simple cyst, benign). **(B)** Axial T2-weighted fat-suppressed MR image shows an oval circumscribed mass with high signal intensity (arrow). **(C)** Axial contrast-enhanced T1-weighted fat-suppressed MR image shows no enhancement of the cyst (arrow).

reverberation echoes are reduced, which allows improved characterization of simple cysts by eliminating artifactual internal echoes often seen in fluid (Fig 3) (15). Additionally, it sharpens the margin of a lesion and accentuates posterior features. Harmonic imaging can be especially helpful in depicting clustered microcysts and excluding a solid mass. However, it has drawbacks such as reduced penetration depth and axial resolution (17,18).

Application of color or power Doppler imaging may increase diagnostic certainty in the evaluation of cystic breast lesions. Although absence of vascularity cannot be used to determine whether an anechoic or hypoechoic mass is a cyst, it may support the diagnosis of a cyst if the mass is anechoic and has a circumscribed margin and an oval shape. Conversely, vascularity within an anechoic mass suggests that the mass is solid and raises suspicion for malignancy. High-grade invasive cancer and metastatic lymph nodes occasionally can appear anechoic, and demonstration of flow within the mass then requires biopsy (Fig 4) (15).

US elastography can improve specificity in the evaluation of cystic breast masses by measuring tissue stiffness with strain or shear wave (Fig 4). A “bull’s eye” artifact has been described as a characteristic feature of benign breast cysts with some strain elastography software. This can appear as

a round or oval lesion with a stiff rim associated with two soft spots—one located centrally and the other posteriorly (19). The bull’s eye artifact can reduce the need for biopsies of benign cystic lesions (19), but it is not observed with the shear-wave elastography technique (20). Elastography can help differentiate circumscribed high-grade invasive ductal carcinomas, which typically show a rim of surrounding stiffness, from complicated cysts, which generally are much softer than malignant lesions (20).

Simple Cysts

The criteria for diagnosing simple cysts on US images are a circumscribed margin, an anechoic round or oval shape, a thin imperceptible wall, and posterior enhancement (Fig 2). The presence of these features establishes the diagnosis of a simple cyst that can be dismissed as a benign finding (BI-RADS category 2). Color or power Doppler imaging applied to the cyst demonstrates no internal vascularity, and flow may be seen in the tissue surrounding the cyst. Thin septations may be present in lobulated simple cysts or when several cysts abut each other. To be defined as a simple cyst, the septations must be less than 0.5 mm in thickness (Fig 5) (6).

Differentiating simple cysts from those with internal echoes or solid masses is sometimes difficult. Posterior enhancement

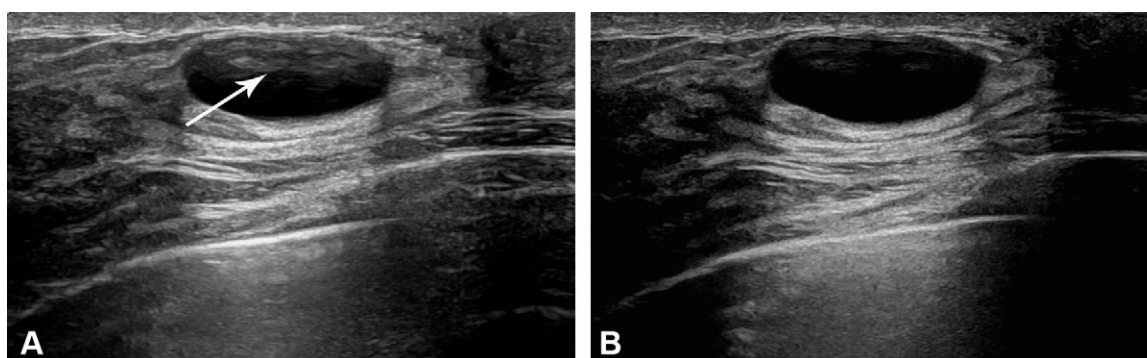


Figure 3. Simple cyst in a 39-year-old woman with a history of contralateral surgery for breast cancer. **(A)** US image shows reverberation artifacts (arrow) in an anterior nondependent portion of a cyst. **(B)** US image using tissue harmonic imaging shows reduced near-field artifactual echoes and intensified posterior features.

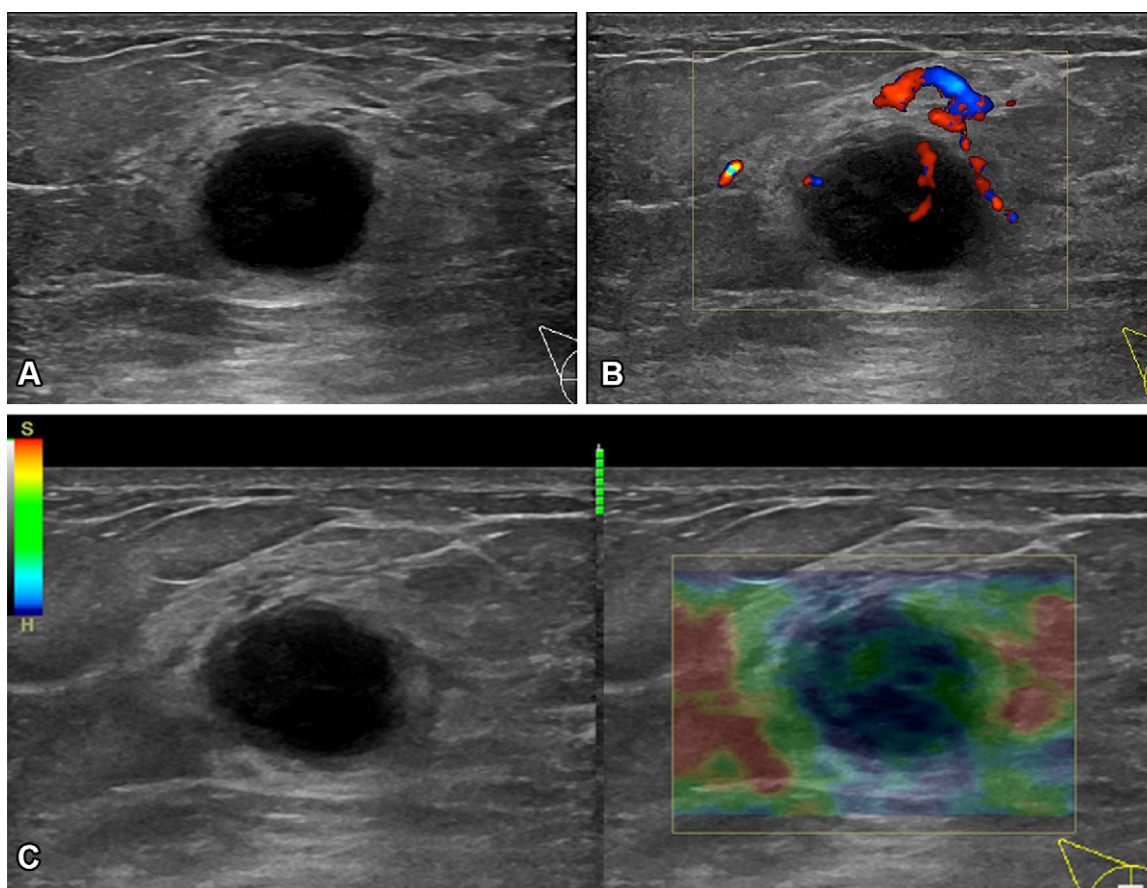


Figure 4. Triple-negative breast cancer (medullary carcinoma) in a 44-year-old woman who presented with a palpable mass. **(A)** US image shows a round mass appearing anechoic. Posterior enhancement also is seen. **(B)** Color Doppler US image shows increased vascularity of the mass, which implies a solid lesion. **(C)** Strain elastogram shows the hardness of the mass, with red indicating soft and blue indicating hard. The mass also shows partially indistinct margins. A US-guided core-needle biopsy was performed, and the result was triple-negative breast cancer (medullary carcinoma).

may be subtle or not always apparent, especially when small cysts are located adjacent to the pectoral muscle. Occasionally, simple cysts smaller than 8 mm are difficult to characterize, especially when located deep in the breast (21,22). Some very hypoechoic solid lesions may appear as anechoic. Additionally, homogeneous solid lesions, including high-grade carcinomas, may exhibit posterior enhancement, which could cause them

to be mistaken for simple cysts. However, most cystic-appearing invasive ductal carcinomas or other malignancies show at least a focally indistinct margin or thickened wall and internal vascularity (Fig 6).

Cysts often are multiple and bilateral. In the American College of Radiology Imaging Network (ACRIN) 6666 protocol for screening US in women with dense breasts and elevated risk,

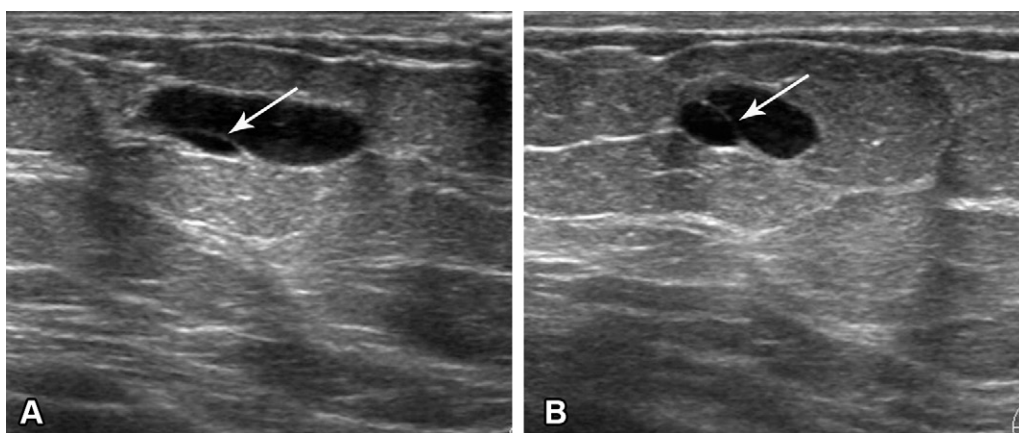


Figure 5. Simple cyst in a 56-year-old woman with a mass at screening mammography. Radial (**A**) and antiradial (**B**) US images show a cyst with a thin (<0.5 mm) septum (arrow).

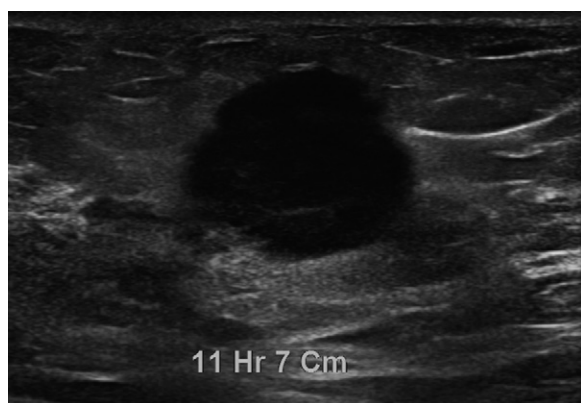


Figure 6. Invasive ductal carcinoma, not otherwise specified type, in a 43-year-old woman who presented with a palpable mass. US image shows a nearly anechoic mass with posterior enhancement that mimics a cyst. However, the entire margin is not circumscribed and shows some indistinct margin. US-guided core-needle biopsy was performed and revealed triple-negative breast cancer (invasive ductal carcinoma).

simple cysts were observed in 998 (37.5%) of 2659 participants in the first round of screening and 1255 (47.1%) of 2662 participants over 3 years. Nearly half (48%) of the participants with simple cysts had cysts in both breasts. Because cysts are hormone sensitive, their size and number fluctuate with the menstrual cycle and are most prominent in the premenstrual phase (4). Interval breast cyst formation is not uncommon, and most cysts disappear naturally. Spontaneous regression of interval benign cysts is seen in more than half of lesions within 1 year (23).

It is unnecessary to report the measurements of every simple cyst. When multiple cysts are present, especially in both breasts, it usually is adequate to document the location and measurements of the largest cyst in each breast (6).

On mammograms, a simple cyst can be seen as a circumscribed mass. Multiple bilateral circumscribed masses (at least three overall and one in each breast) on screening mammograms are commonly caused by simple cysts and can be dismissed as benign findings (24). Furthermore, fluctuation

in size may be seen at serial examination (Fig 7). Calcifications in the walls of cysts can be seen as rim calcifications, whereas sedimented calcifications in macro- or microcysts can be seen as layering (6).

Simple cysts are common incidental findings at breast MRI performed for high-risk screening, to determine the extent of disease in newly diagnosed breast cancer, or to evaluate the response to neoadjuvant therapy. These cysts usually are circumscribed round or oval fluid-filled masses with an imperceptible wall. They typically exhibit high signal intensity on T2-weighted images, which indicates fluid within the cyst, and low signal intensity on T1-weighted images. There is no enhancement following administration of contrast agent (Fig 2). Although rim enhancement can be suggestive of malignancy, a cyst can enhance peripherally and show rim enhancement if there is surrounding pericystic inflammation. However, it is usually thin, uniform, and hyperintense on T2-weighted images, which confirms the cystic nature (Fig 8).

Complicated Cysts

Complicated cysts are cysts that contain debris, including proteins from cell turnover, milk, fat, pus, and blood (4). A galactocele is a cystically dilated terminal duct and terminal ductule filled with milk. It most commonly occurs during lactation and occasionally in the third trimester of pregnancy. The radiologic appearance of a galactocele depends on the amount of fat and protein within the milk (25).

Complicated cysts are masses that otherwise meet the sonographic criteria for simple cysts, except they are not anechoic on US images. A lesion can easily be diagnosed as a complicated cyst with debris and categorized as BI-RADS category 2 in cases of mobile internal echoes, a fluid-debris level, or echogenic reflector scintillating during real-time US image evaluation (Figs 9, 10). Otherwise, a complicated cyst should show internal homogeneous low-level echogenicity without mural nodules, thick septations, thick walls, or any other suggestion of a solid component (Fig 10) (6). The presence of echoes that are caused by debris or hemorrhage associated with vascularity within a complicated cyst, any thickening or irregularity of the wall, or any change in size at follow-up may require further investigation to exclude malignancy (1,7).

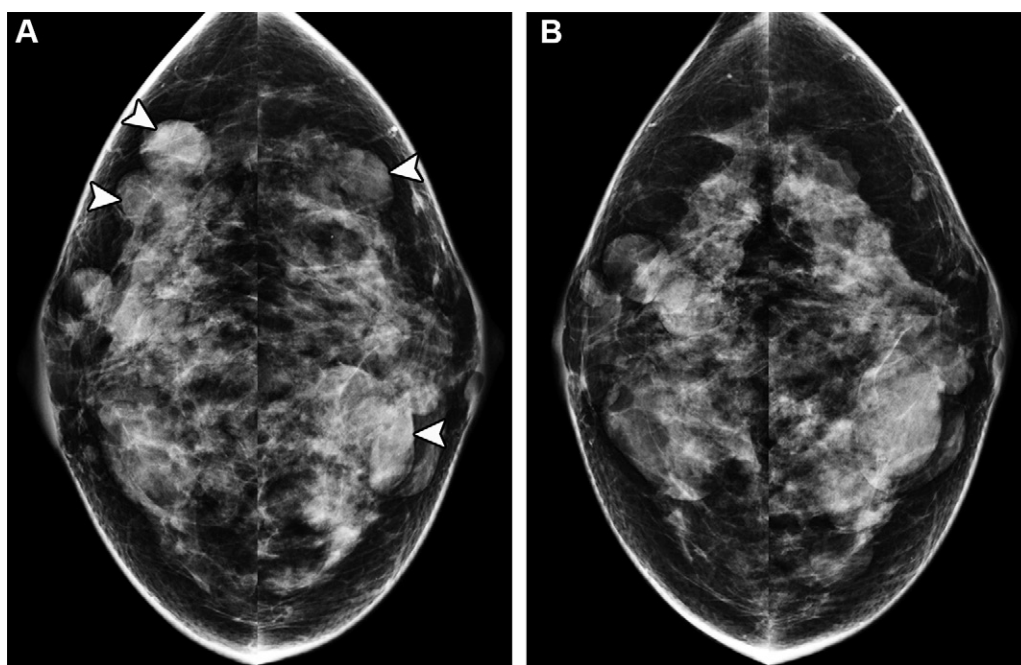


Figure 7. Multiple cysts in a 44-year-old woman. **(A)** Craniocaudal mammograms show multiple bilateral masses with obscured margins (arrowheads), which were confirmed as multiple simple cysts at US (not shown). **(B)** Craniocaudal mammograms obtained 1 year later show fluctuation in the size and number of masses, with some decrease in size.

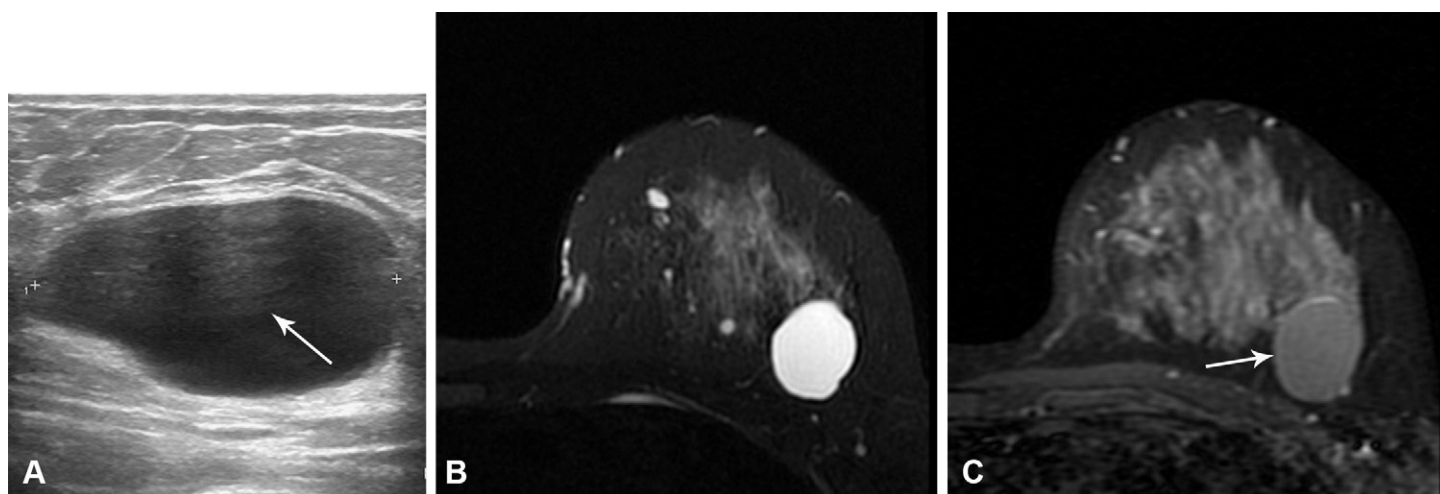


Figure 8. Simple cyst in a 43-year-old woman. **(A)** US image shows a simple cyst with reverberation artifacts (arrow) paralleling the anterior wall of the cyst. Breast MRI was performed to evaluate the disease extent of breast cancer (not shown). **(B)** Axial T2-weighted fat-suppressed MR image shows an oval circumscribed mass with high signal intensity. **(C)** Axial contrast-enhanced T1-weighted fat-suppressed subtraction MR image shows thin and uniform enhancement of the cyst (arrow). Occasionally, cysts can enhance peripherally, which also is a benign finding if the cyst is thin and uniform.

Complicated cysts with debris must be distinguished from complex cystic and solid masses categorized as BI-RADS category 4, suspicious findings.

On mammograms, complicated cysts are visualized as solitary or multiple round or oval bilateral masses (Fig 10) with possible layering calcifications, as seen in simple cysts. Rim calcification sometimes is evident and indicates benignity (4).

On breast MR images, complicated cysts usually appear as circumscribed round or oval fluid-filled masses with an imperceptible wall and show variable signal intensity characteristics, depending on their contents. Hemorrhagic and proteinaceous complicated cysts appear hyperintense on T1-weighted images and hypointense on T2-weighted images (Fig 11).

Complicated cysts frequently are asymptomatic and often accompany simple cysts. According to the ACRIN 6666 trial, 376 of 2662 (14.1%) women had complicated cysts; of these, 301 (80.1%) women also had at least one simple cyst, and 84 (22.3%) had multiple bilateral complicated cysts (4). The latter (at least three overall and one in each breast), or complicated cysts combined with multiple simple cysts at US, can be assessed as BI-RADS category 2 with a recommendation for routine follow-up. Because complicated cysts often manifest as homogeneous low-level echoes with an imperceptible wall, they can be challenging to distinguish from a solid mass. However, multiple bilateral circumscribed masses (ie, complicated cysts or solid oval circumscribed masses) detected at screening US are benign (26). Multiple bilateral similarly

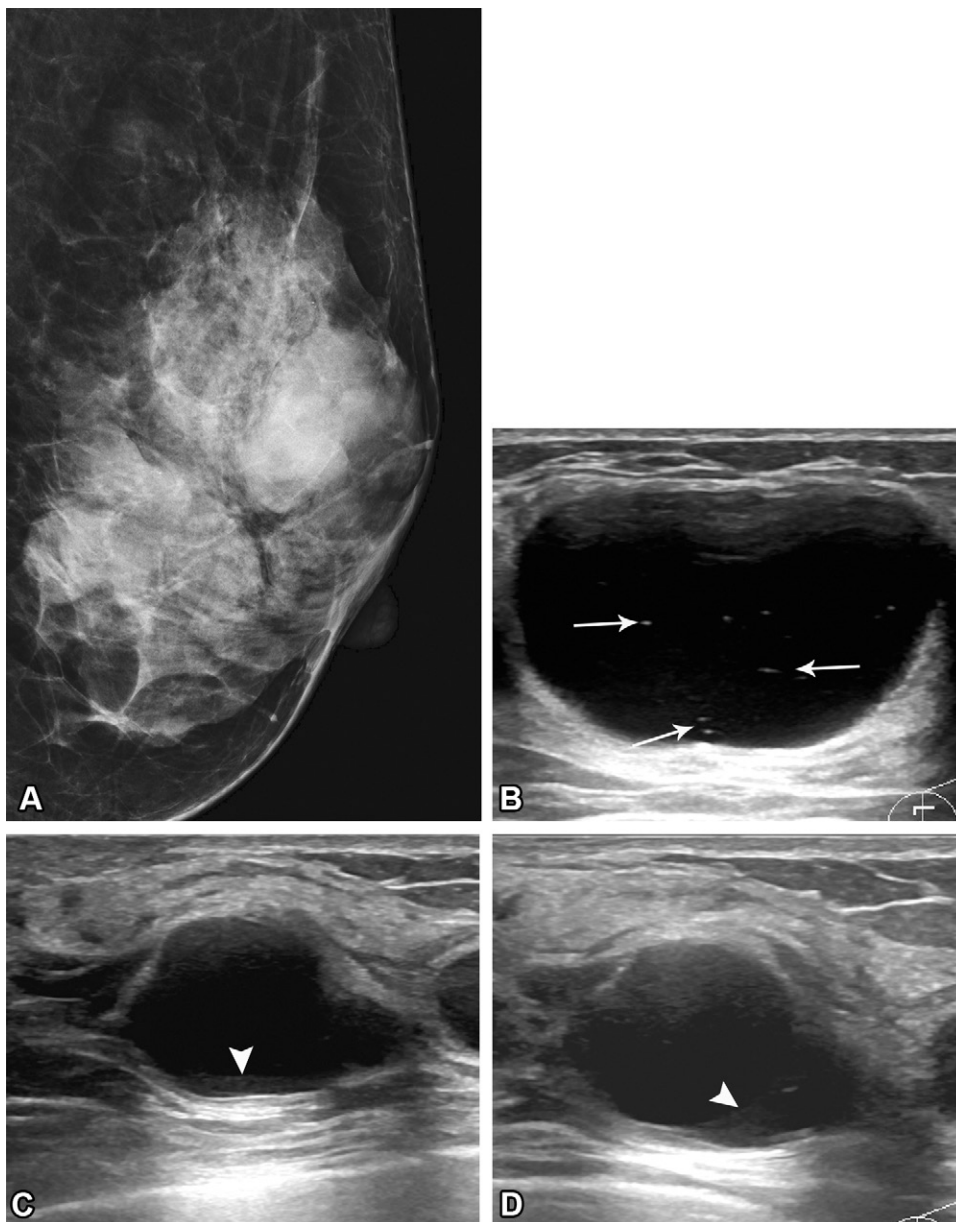


Figure 9. Complicated cysts in a 49-year-old woman who presented with pain. **(A)** Mediolateral oblique mammogram shows masses with obscured margins in the left breast. US was performed for further evaluation because of a history of right nipple-sparing mastectomy for breast cancer. **(B)** US image shows an oval circumscribed mass with posterior enhancement and echogenic reflectors (arrows) that were mobile at real-time evaluation. **(C, D)** US images show an oval circumscribed mass with posterior enhancement and fluid-debris level (arrowhead) that shifted according to position change. Several simple cysts were also seen in the left breast (not shown). Fluid-debris level or mobile internal echoes at real-time US evaluation are pathognomonic findings of a complicated cyst.

appearing circumscribed masses at US can be assessed as benign with follow-up in 1 year, as for screening mammography (24). However, careful evaluation of each mass is important because cancer can coexist with multiple bilateral circumscribed benign masses (Fig S3) (27).

The reported malignancy rate of complicated cysts was 0.3% (four of 1526) across eight prior series (Table 1) (1,2,4,7,28–31). Therefore, complicated cysts with debris can be assessed as BI-RADS category 2 (32). Aujero et al (7) reported that all 183 complicated cysts were benign in 114 postmenopausal women. The study supported follow-up imaging, rather than aspiration or biopsy, for asymptomatic complicated cysts, regardless of patient age (7).

An isolated complicated cyst with uniform low-level echoes can mimic a solid oval circumscribed mass, particularly if the lesion is very small or deeply located. However, in prior studies (29–31), aspiration showed that 12.4% (64 of 517) of masses thought to be complicated cysts were solid masses. An isolated

complicated cyst with uniform low-level echoes can be categorized as BI-RADS category 3, probably benign (33). Color or power Doppler US imaging may be helpful and increase diagnostic certainty in such cases.

Clustered Microcysts

Clustered microcysts, part of the spectrum of benign cystic changes, are cystic dilatations of individual acini within the terminal duct lobular unit. Tissue diagnoses associated with clustered microcysts include fibrocystic changes, apocrine metaplasia, and benign cysts (6,10,34).

Clustered microcysts are defined by BI-RADS as lesions comprising a cluster of anechoic masses, each smaller than 2–3 mm, with thin (<0.5 mm) septations and no discrete solid component at breast US (Fig 12) (6). However, some studies included lesions with individual microcysts up to 5–7 mm in size and suggested broadening the definition to include lesions with larger cysts (4,9,34). The margin should be circumscribed

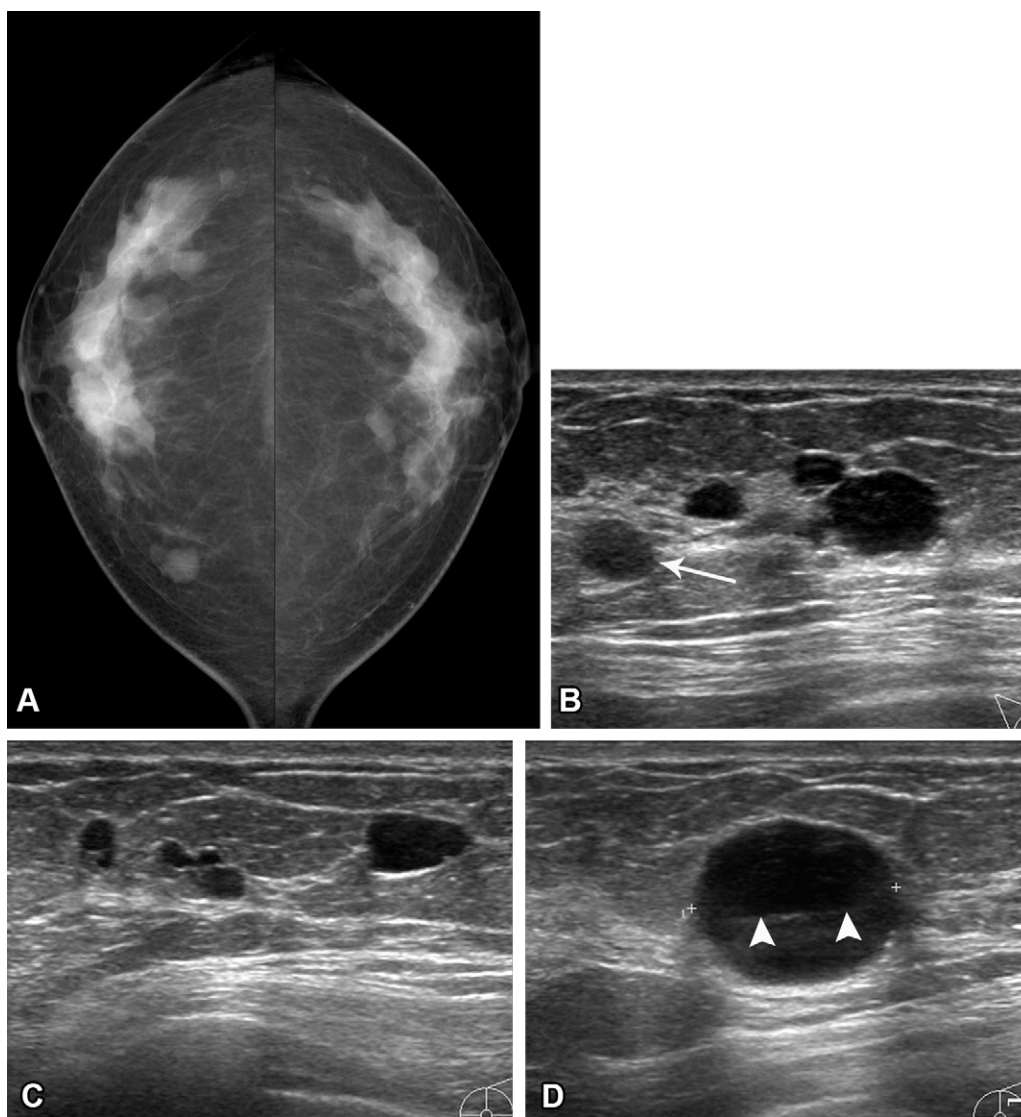


Figure 10. Complicated cyst in a 46-year-old woman with masses at screening mammography. **(A)** Bilateral craniocaudal mammograms show multiple bilateral masses with obscured margins in both breasts. **(B, C)** US images show multiple oval circumscribed anechoic masses with posterior enhancement (multiple simple cysts), as well as an adjacent oval circumscribed mass with homogeneous low-level echoes and posterior enhancement (complicated cyst, arrow in **B**). **(D)** US image shows an oval circumscribed mass with posterior enhancement and fluid-debris level (arrowheads), which is typical of a complicated cyst. At US, multiple bilateral complicated cysts, or a complicated cyst combined with multiple simple cysts (at least three overall and one in each breast), can be assessed as BI-RADS category 2, benign.

and fluid-debris levels seen within individual microcysts (34). At mammography, clustered microcysts can manifest as an oval or round mass with a circumscribed or microlobulated margin (9,13).

Clustered microcysts are common findings in women of all ages, most often in perimenopausal women, and are reported in up to 6% of all US examinations (34). According to the ACRIN 6666 trial, 104 of 2662 (3.9%) women had clustered microcysts with associated simple and complicated cysts (4). During follow-up, clustered microcysts were stable in most women (116 of 158 [73.4%]) or either spontaneously resolved or shrank in 41 of 158 (25.9%) women in a recent study by Goldbach et al (10) (Fig 13).

The assessment and management recommendations for clustered microcysts may be confusing, even for experienced radiologists. In the multicenter ACRIN 6666 protocol, which began in 2004, clustered microcysts were included in the prospective criteria for BI-RADS category 3 (35). According to BI-RADS, clustered microcysts can be assessed as category 2 when clearly composed of simple cysts (6). However, imaging surveillance and a BI-RADS category 3 assessment may be appropriate for smaller or deeply located clustered microcysts. Based on

prior studies with one malignancy reported among 216 (0.5%) cases, these lesions have reduced diagnostic certainty. BI-RADS also states, “The relatively small number of cases studied limits precision in estimating the likelihood of malignancy to be $\leq 2\%$; the data would be more convincing if at least 500 cases were studied” (6). Since then, two additional studies on the outcomes of clustered microcysts have been published, adding to the number of cases. Goldbach et al (10) reported the outcomes of 196 clustered microcysts, which showed no malignancies after at least 24 months of imaging follow-up or tissue sampling. Greenwood et al (9) also found no malignancy in 95 clustered microcysts. Tanaka et al (8) reported outcomes of 52 lesions categorized as clustered microcysts, with 8 malignancies out of 52 (8). However, strict criteria of clustered microcysts probably were not applied because intervening or associated solid components were seen in lesions illustrated in the study.

Excluding the study by Tanaka et al (8), seven studies investigated a combined total of 526 clustered microcysts (Table 2) (1,2,4,9,10,31,34). The malignancy rate among these microcysts was extremely low, with one malignancy in 526 lesions (0.2%). One malignancy in 123 lesions (0.8%) was detected during the second screening in the ACRIN 6666 study

Figure 11. Complicated cysts in a 47-year-old woman with invasive ductal carcinoma in the left breast. Multiple simple and complicated cysts in both breasts were seen at whole breast US (not shown). **(A)** Axial T2-weighted fat-suppressed MR image shows multiple oval or round circumscribed masses with high signal intensity (arrow), which indicates fluid-filled masses. **(B)** Axial T1-weighted fat-suppressed MR image shows varying signal intensity of the masses, specifically two abutting cysts (arrow, arrowhead). Note the different signal intensities of the two cysts on the T1- and T2-weighted images in **A** and **B**. **(C)** Axial contrast-enhanced T1-weighted fat-suppressed subtraction MR image shows no enhancement (arrow, arrowhead) of multiple simple and complicated cysts in both breasts. **(D)** US image shows corresponding two abutting complicated cysts (arrow, arrowhead).

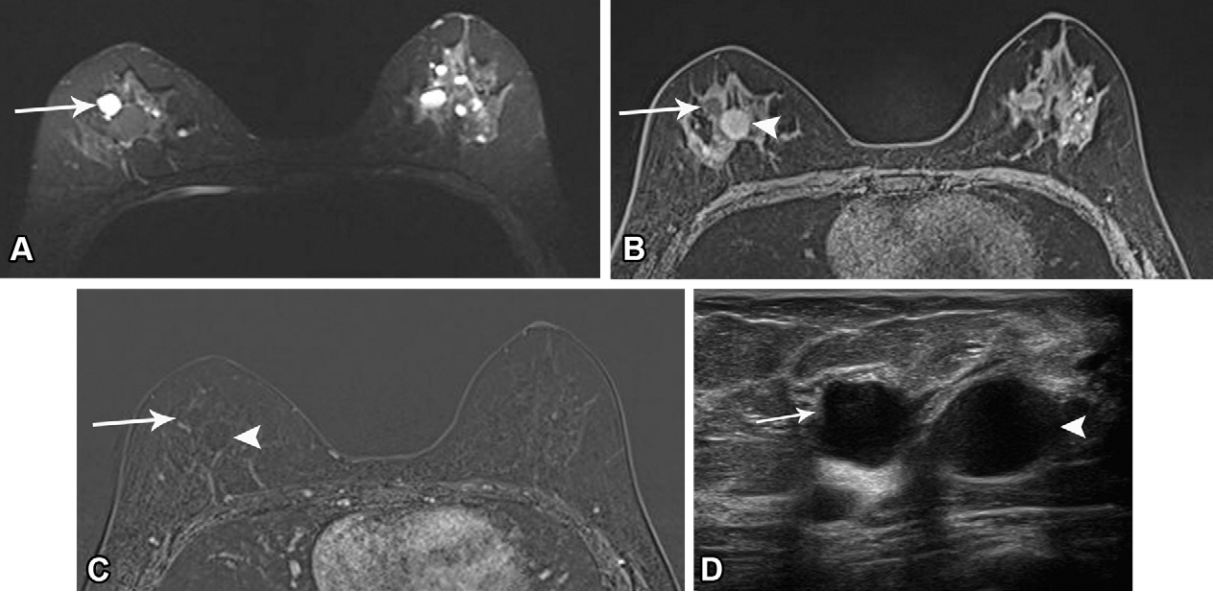


Table 1: Reported Outcomes of Complicated Cysts

Study and Year*	No. of Complicated Cysts	No. of Lesions Aspirated or Biopsied	No. of Malignancies	Details of Malignancy
Kolb et al (28), 1998	126	30	0	NA
Venta et al (29), 1999	308	123	1 (0.3%)	3-mm DCIS in a papilloma
Buchberger et al (30), 1999	133	133	0	NA
Berg et al (1), 2003	38	38	0	NA
Chang et al (2), 2007	35	35	0	NA
Daly et al (31), 2008	228	228	1 (0.4%)	4-mm IDC NOS
Berg et al (4), 2010	475	74	2 (0.4%)	1 6-mm ILC, 1 IDC-DCIS
Aujero et al (7), 2019	183	85	0	NA
Totals for all studies	1526	746	4 (0.3%)	...

Note.—DCIS = ductal carcinoma in situ, IDC = invasive ductal carcinoma, ILC = invasive lobular carcinoma, NA = not applicable, NOS = not otherwise specified.

* Numbers in parentheses are reference numbers.

(4). The lesion was initially described as solitary clustered microcysts and then as a complex cystic and solid mass at follow-up. However, the authors noted that it might not have been the same lesion initially classified as clustered microcysts. They assumed that the original clustered microcysts resolved and a new lesion formed (4). Based on the studies currently reported, the risk of malignancy for clustered microcysts is extremely low. Therefore, BI-RADS category 2 is most appropriate for clustered microcysts that have a typical sonographic appearance (10,32). The presence of indistinct or angular margins, discrete solid components, associated

suspicious calcifications at mammography, or a suspicious change at subsequent imaging should prompt a BI-RADS category 4 assessment and a recommendation for biopsy (Fig 14) (36).

Individual microcysts may demonstrate low-level echoes, which complicates the exclusion of a solid component. In such cases, harmonic imaging may be helpful to increase diagnostic certainty. If there is uncertainty, such as very small or deeply located clustered microcysts (eg, greater than 3 cm from the skin at US), BI-RADS category 3, with a 6-month follow-up, is reasonable.

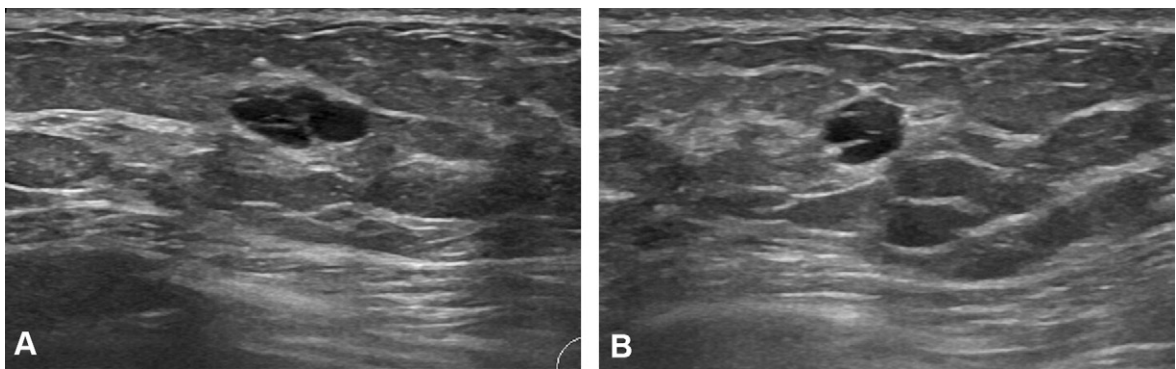


Figure 12. Clustered microcysts in a 52-year-old woman. Radial **(A)** and antiradial **(B)** US images show clustered microcysts as small anechoic cysts with thin septations in an oval circumscribed lesion. No solid component is present in any of the cysts.

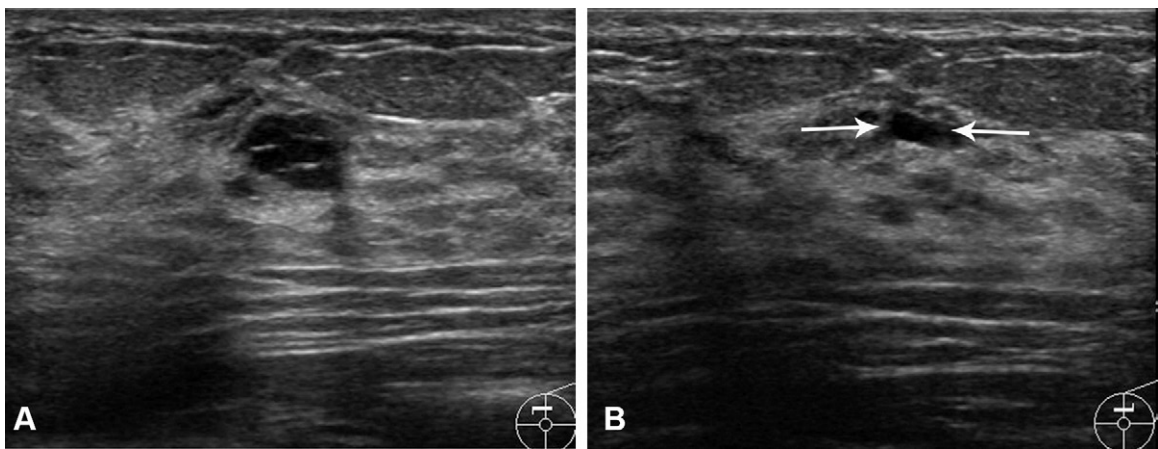


Figure 13. Clustered microcysts in a 24-year-old woman with breast pain. **(A)** US image shows clustered microcysts as 2–3-mm cysts and thin septations in an oval anechoic lesion in the left breast at the 12-o’clock position. **(B)** US image at 6-month follow-up shows that the lesion (arrows) decreased in size.

Table 2: Reported Outcomes of Clustered Microcysts				
Study and Year*	No. of Clustered Microcysts	No. of Lesions Aspirated or Biopsied	No. of Malignancies	Details of Malignancy
Berg et al (1), 2003	16	16	0	NA
Berg (34), 2005	66	13	0	NA
Chang et al (2), 2007	15	15	0	NA
Daly et al (31), 2008	15	15	0	NA
Berg et al (4), 2010	123	5	1 (0.8%)	18-mm ILC
Greenwood et al (9), 2017	95	16	0	NA
Goldbach et al (10), 2020	196	38	0	NA
Totals for all studies	526	118	1 (0.2%)	...

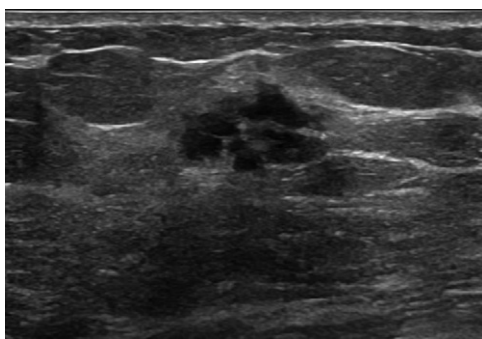
Note.—ILC = invasive lobular carcinoma, NA = not applicable.
* Numbers in parentheses are reference numbers.

Complex Cystic and Solid Masses

According to the BI-RADS criteria, the echo pattern of “complex cystic and solid” indicates a complex mass containing both anechoic (cystic or fluid) and echogenic (solid) components at breast US (6). Lesions found in complex cystic and solid masses may have one or all of the following characteristics: a thick wall

(≥0.5 mm) (Fig 15), thick internal septations (≥0.5 mm) (Fig 16), an intracystic or mural mass (Fig 17), and both solid and cystic components (Fig 18) (4). Although echogenicity alone has low specificity in evaluation of a breast mass, complex cystic and solid masses usually are evaluated as BI-RADS category 4 and accompanied by a recommendation for biopsy.

Figure 14. Invasive ductal carcinoma in a 58-year-old woman who presented with a palpable breast mass. US image shows an irregular mass composed of multiple anechoic or hypoechoic lesions. There are some thick internal septations, as well as a suspected solid component within the lesion. This should be considered as a complex cystic and solid mass. It was deemed suspicious, and a US-guided core-needle biopsy was performed. The biopsy result was invasive ductal carcinoma, not otherwise specified.



The reported malignancy rate of complex cystic and solid masses is 22%–43% (1,4,5). These lesions can be associated with various benign and malignant breast diseases. Benign causes of thick-walled cystic lesions, with or without thick internal septations, are ruptured or inflamed cysts and fibrocystic changes, especially with apocrine metaplasia, abscesses, fat necrosis, seromas, and hematomas (1,4,5). For probably benign causes of complex cystic and solid masses, follow-up US to resolution is recommended to ensure accurate assessment and appropriate management while carefully ruling out underlying malignancy. The sonographic appearances of fibrocystic changes vary with the relative degrees of cystic dilatation and sclerosis. When the cystic change predominates, sonographic diagnosis is easy. However, fibrocystic changes can manifest as a complex cystic and solid appearance when sclerosis and papillary apocrine metaplasia predominate.

Mastitis causes diffuse changes in the sonographic appearance of breast tissue, such as disrupted normal architecture, decreased parenchymal echogenicity caused by edema, dilated and thickened ducts, and skin thickening. With an abscess, a poorly defined anechoic or hypoechoic fluid-filled space or a complex cystic lesion can be seen (Fig 15) (37). US findings associated with clinical symptoms (ie, local heat, swelling, and redness of the skin overlying the breast) are highly suggestive of an abscess. A chronic abscess may contain sufficient granulation tissue and fibrosis, develop into a complex cystic and solid mass, or have a solid appearance at US.

Fat necrosis and hematoma can manifest as a thick-walled cystic lesion or a complex cystic and solid mass. A hematoma or fat necrosis may be suspected in appropriate clinical settings, such as in cases of recent trauma, needle biopsy, or surgery. Correlation with mammography is mandatory, and follow-up US can be performed instead of needle biopsy if the mass shows characteristically benign mammographic findings (Fig 19). After surgery, most seromas appear as simple or complicated cysts with diffuse low-level internal echoes. A complex cystic and solid mass at the lumpectomy site can be diagnosed definitively based on US findings because the postoperative fluid collection corresponds directly with the lumpectomy site (Fig S4).

Both benign and malignant papillary lesions often manifest as intraductal or intracystic masses. Intraductal papillomas are characterized by fibrovascular cores and epithelium, with an inner myoepithelial and outer columnar epithelial cell layer growing in an arborescent frondlike pattern with-

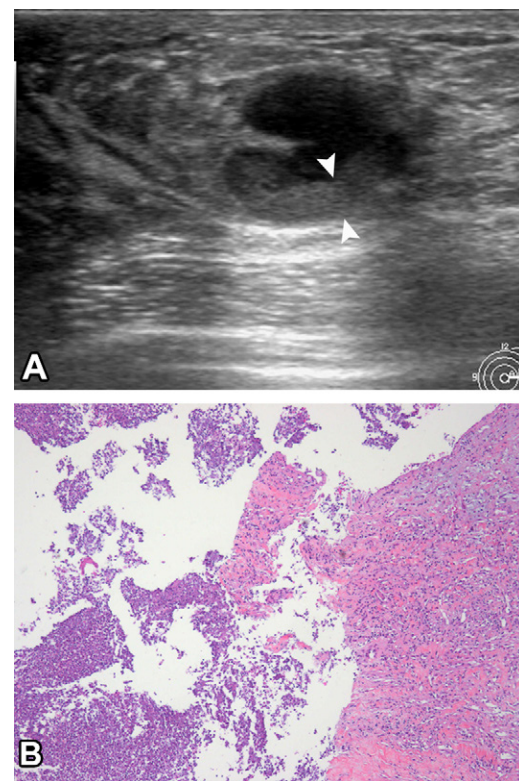


Figure 15. Abscess in a 38-year-old woman who presented with a palpable breast mass. (A) US image shows a complex cystic and solid mass with an eccentric thickened wall (arrowheads) in the palpable area. A US-guided core-needle biopsy was performed. (B) Photomicrograph of histopathologic specimen shows inflammatory infiltrate, fibrosis, foam cells (macrophages), inflammatory giant cells, and epithelial degeneration. (H-E stain; original magnification, $\times 40$.)

in the duct. Papillomas tend to secrete fluid into the ducts, which leads to duct dilatation and nipple discharge. Hypersecretion and duct obstruction result in cystic dilation of the duct around the papilloma. Papillomas usually appear as circumscribed oval hypoechoic intraductal (or intracystic) masses on US images, with blood flow to a vascular stalk seen on Doppler US images (38). Atypical epithelial proliferation, such as atypical ductal hyperplasia and ductal carcinoma in situ, may occur in papillomas and usually is of low grade (39).

Intraductal papillary carcinoma is a malignant neoplastic epithelial proliferation with papillary architectural features occurring in the lumen of the ductal-lobular system. It can be differentiated from intraductal papilloma by the lack of myoepithelial cell proliferation (40). Generally, intraductal papillary carcinoma appears as a complex cystic and solid mass with a solid vascularized component (Fig S5). Larger irregular intracystic lesions with indistinct margins and markedly mural vascularized nodules suggest a probable invasive form, whereas circumscribed cystic lesions with a smaller solid nodule suggest a noninvasive form (41).

Malignant breast masses can be seen as complex cystic and solid masses at US if there are necrotic or hemorrhagic components within the mass. A rapidly growing invasive carcinoma

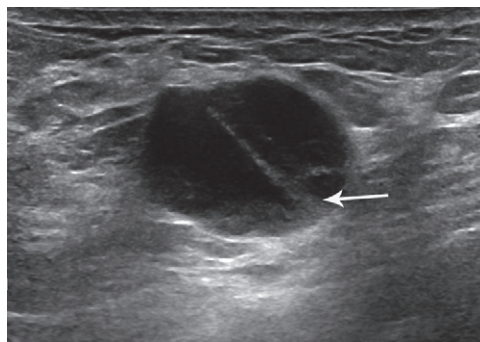


Figure 16. Encapsulated papillary carcinoma with invasion in a 56-year-old woman who presented with a palpable breast mass. US image shows a complex cystic and solid mass with a thickened wall and septations (≥ 0.5 mm) (arrow). A US-guided core-needle biopsy was performed at an outside clinic and revealed papilloma with atypical ductal hyperplasia. The final pathologic diagnosis after excision was encapsulated papillary carcinoma with invasion.

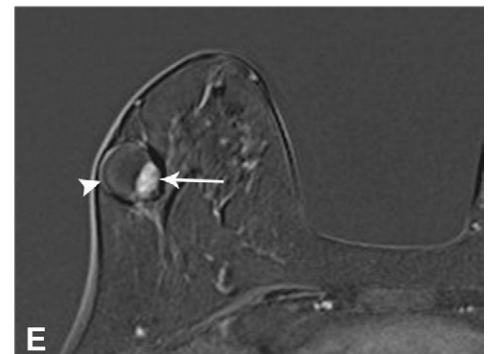
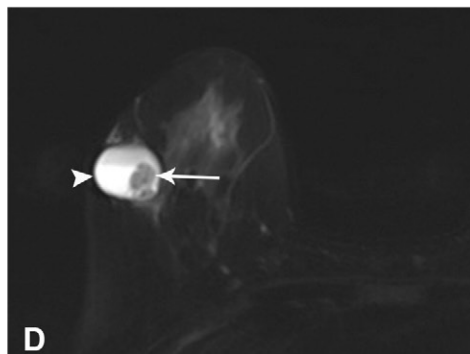
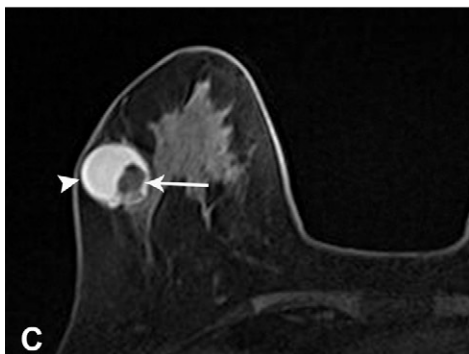
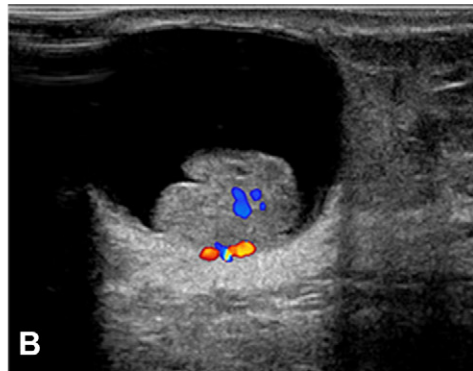
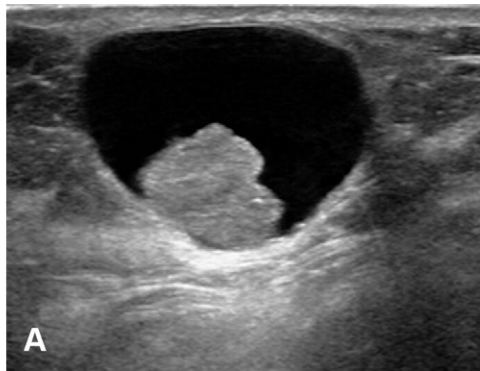


Figure 17. Ductal carcinoma in situ arising in a papilloma in a 50-year-old woman who presented with a palpable mass. **(A)** US image shows a complex cystic and solid mass in the palpable area of the right breast. **(B)** Color Doppler US image shows increased vascularity of the solid portion. A US-guided core-needle biopsy targeting the solid mural nodule revealed a low-grade ductal carcinoma in situ. **(C–E)** Axial T1-weighted fat-suppressed **(C)**, T2-weighted fat-suppressed **(D)**, and contrast-enhanced T1-weighted fat-suppressed subtraction **(E)** MR images show a round mass with enhancing solid (arrow) and hemorrhagic cystic (arrowhead) portions. Breast-conserving surgery was performed, and the tumor with a hemorrhagic cyst was seen. The pathologic diagnosis was ductal carcinoma in situ arising in the papilloma.

that most often is a poorly differentiated invasive ductal carcinoma, medullary carcinoma, and other malignancies may be seen as complex cystic and solid masses (2,3). The sonographic findings vary greatly and reflect a gross morphologic and histologic heterogeneity. On US images, tumor necrosis usually is seen as areas of cystic or hemorrhagic changes inside the solid mass (Fig 18). Necrosis is common in high-grade carcinomas and very unusual in low-grade carcinomas (4). Ductal carcinoma in situ is a heterogeneous disease entity with a variable appearance at US. A complex echotexture also can be present and probably is caused by the lobular portion distention of the terminal ductal lobular unit (42).

Assessment and Management

Figure 20 summarizes the diagnostic approach and BI-RADS category assessment of cystic breast lesions at US.

Simple cysts can be dismissed as benign (BI-RADS category 2) with a recommendation for routine screening. If the patient is symptomatic with pain or a very large cyst, therapeutic cyst aspiration can be performed, a seemingly discordant clinical scenario (6). Aspiration also can be performed for symptomatic (ie, tender or palpable) complicated cysts or for diagnostic uncertainty of some complicated cysts.

Complex cystic and solid masses are usually considered suspicious findings and require biopsy for definitive diagnosis. Short-term follow-up US of a thick-walled cystic lesion can be performed in patients with a history of trauma or signs of infection. However, if there is any enlargement of such a lesion, a biopsy should be recommended because an underlying malignancy might bleed from trauma or cause duct obstruction and infection (1).

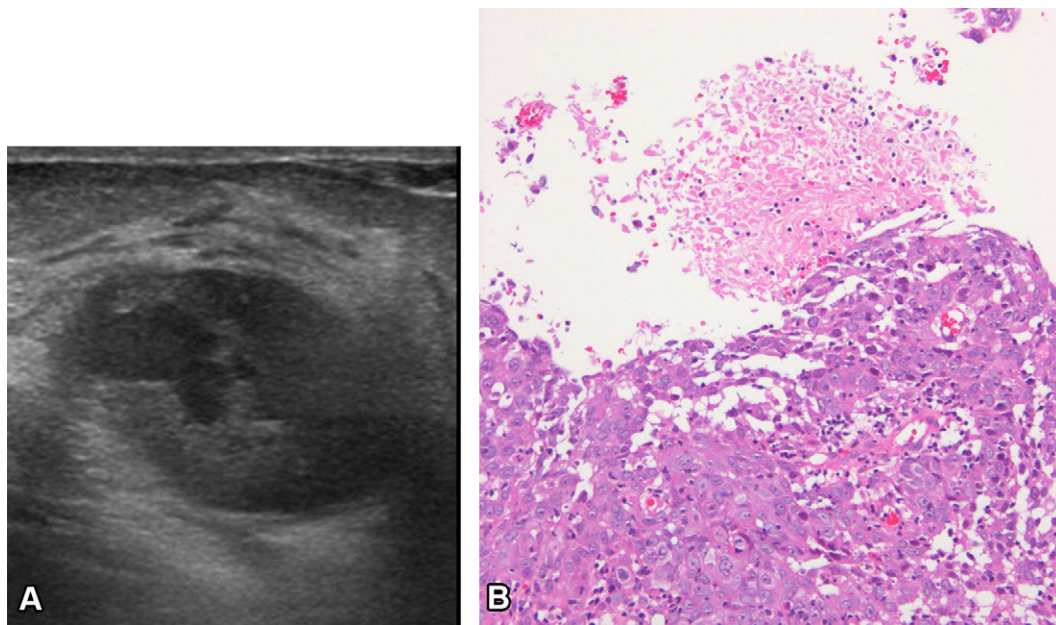


Figure 18. Invasive ductal carcinoma, not otherwise specified, in a 45-year-old woman who presented with a palpable mass. **(A)** US image shows an oval complex cystic and solid mass. **(B)** Photomicrograph of the histopathologic specimen shows invasive ductal carcinoma with central necrosis. (H-E stain; original magnification, $\times 100$.)

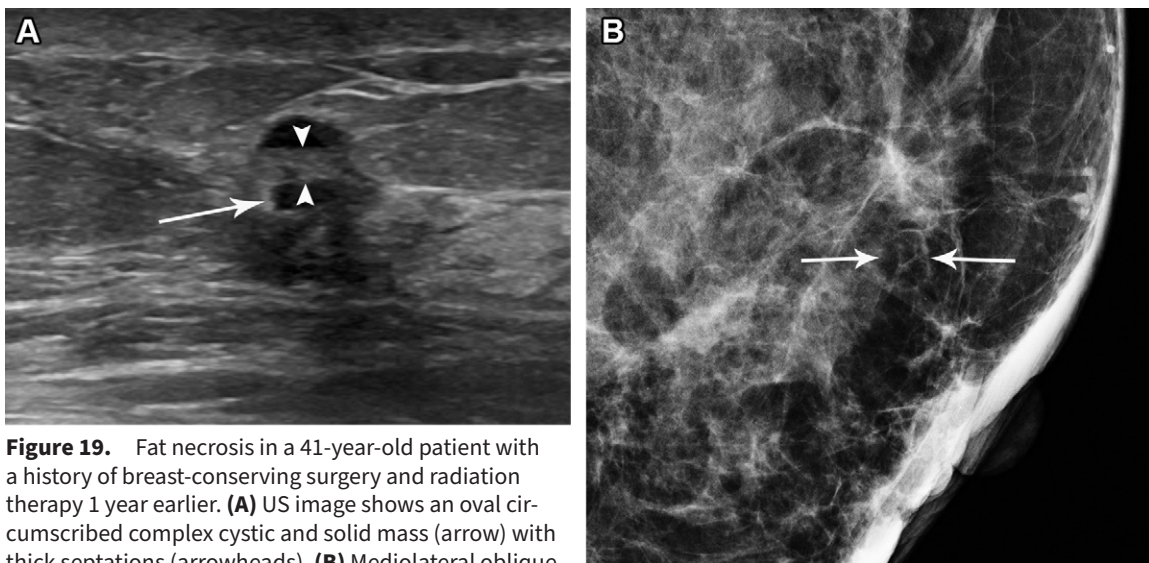


Figure 19. Fat necrosis in a 41-year-old patient with a history of breast-conserving surgery and radiation therapy 1 year earlier. **(A)** US image shows an oval circumscribed complex cystic and solid mass (arrow) with thick septations (arrowheads). **(B)** Mediolateral oblique mammogram shows a circumscribed fat-density mass (arrows) with characteristic benign findings. Because of the clinical history and mammographic findings, no further intervention was needed.

Fine-Needle Aspiration

US-guided fine-needle aspiration is an easy method to assess the nature of cyst contents. The color of breast cyst fluid ranges from white and yellow through brown and gray to blue and black (1). The consistency also can vary from thin and watery to thick and viscid (12). When the cystic fluid is typically characteristic of a benign cyst, such as cloudy yellow or greenish black in color, it can be safely discarded (Fig S6). Routine cytologic examination is deemed unnecessary if the fluid obtained through radiologically guided aspiration is not bloody and if there is complete resolution of the cyst following aspiration. Sending typical cyst fluid

for cytologic examination may yield a false-positive result, which potentially leads to unnecessary additional treatment (4,43). According to a report by Daly et al (31) on complicated cysts, only one of 184 lesions with nonbloody cystic fluid showed atypia at cytologic examination and was confirmed to be fibrocystic disease after surgery. If bloody, the fluid should be sent for cytologic examination. A clip should be placed internally at the aspiration site if there is no residual mass after aspiration (3). This ensures precise localization for future diagnostic or surgical procedures. If the aspirate appears purulent, it should be sent for microbiologic analysis, including culture and staining (1).

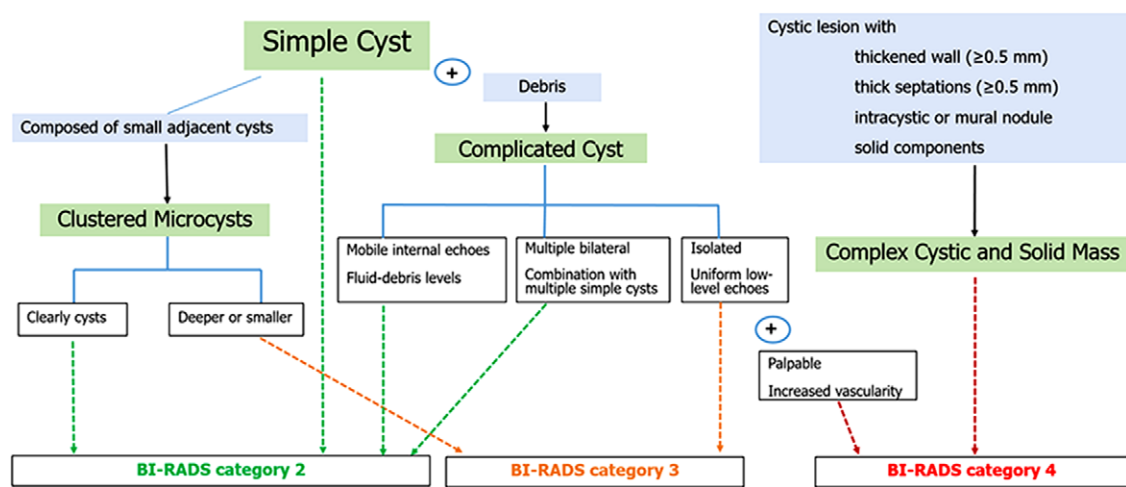


Figure 20. Flow-chart shows the diagnostic approach to, and BI-RADS assessment of, cystic breast lesions on US images.

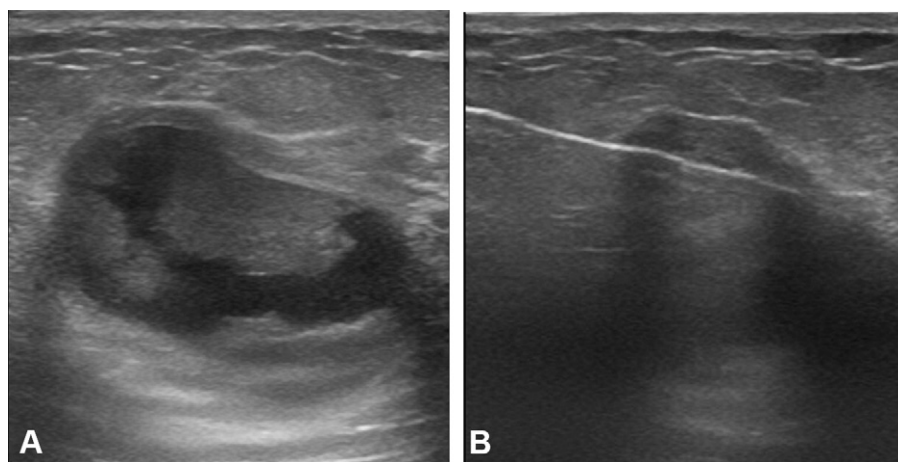


Figure 21. Ductal carcinoma in situ in a 52-year-old woman who presented with a palpable mass. **(A)** US image shows a complex cystic and solid mass, assessed as suspicious (BI-RADS category 4). US-guided biopsy was recommended. **(B)** US image obtained during core-needle biopsy shows sampling of the solid component. The biopsy result was papillary carcinoma in situ.

US-guided Core-Needle Biopsy

Core-needle biopsy under US guidance is an effective method for pathologic diagnosis. It is important to obtain a sample of the solid components during biopsy because the cystic component could actually be necrosis (Fig 21) (3). After biopsy, pathologic findings should be correlated with imaging features to determine concordance. Discordant pathologic findings should be discussed with the pathologist, and a repeat core-needle biopsy or surgical biopsy should be considered (44). Although there can be considerable overlap in US findings between benign and malignant complex cystic and solid masses, the appearance of malignant lesions usually reflects their malignant nature and may involve angular, indistinct, or microlobulated margins or irregular thick walls. The prevalence of malignancy is higher in masses containing mixed cystic and solid components (ie, a cystic portion occupying at least 50% of the mass or a predominantly solid mass) than among masses with a thick wall, thick septations, or cystic lesion with discrete solid mural lesions within a cyst (45,46).

Conclusion

Cystic breast masses are not uncommon findings at mammography, US, and breast MRI. For accurate classification and assessment of cystic breast lesions, it is necessary to obtain US images using a meticulous technique. Simple cysts can be dis-

missed as benign. Complicated cysts with debris and clustered microcysts that meet strict diagnostic criteria can be considered benign, BI-RADS category 2. A BI-RADS category 3 assessment is reasonable in cases of diagnostic uncertainty, even after a thorough US examination. When a cystic lesion shows a thick wall (≥ 0.5 mm), thick internal septations (≥ 0.5 mm), an intracystic mass, or both cystic and solid components, it should be diagnosed as a complex cystic and solid mass. Although follow-up US could be performed if there is a history of trauma or surgery with characteristically benign mammographic findings, complex cystic and solid masses are usually assessed as suspicious, BI-RADS category 4, and require a biopsy. A comprehensive understanding of the imaging findings and management of cystic breast lesions is essential to reduce unnecessary imaging follow-up or biopsies so that cancers are not missed.

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